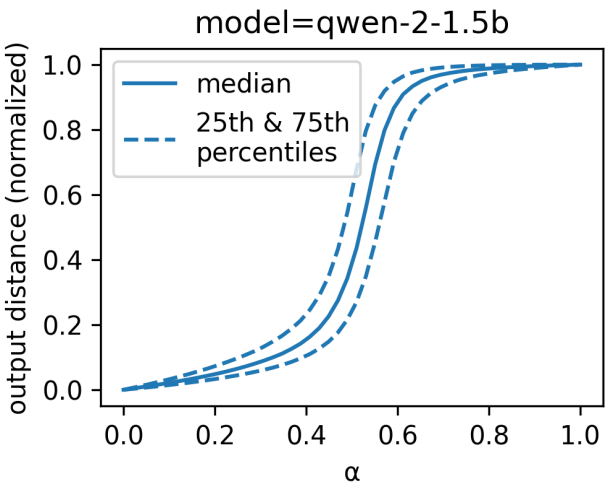

Figure 23: Interpolating and patching after 1st layer

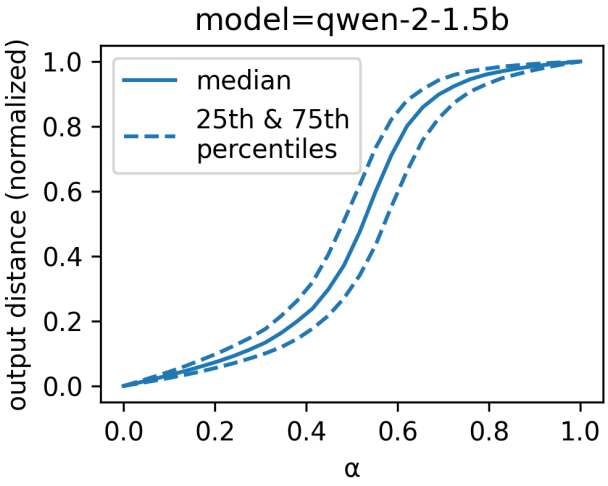

Figure 24: Interpolating and patching after 7th layer

F Direct comparison with polytopes

To directly compare the size of stable regions with polytopes, we analyze gate activations during interpolation between prompts in `Qwen2-0.5B`. For each pair of prompts (p_A, p_B), we collect gate activations at the last layer and count how many gates switch their sign (from positive to negative or vice versa) the beginning ($\alpha = 0$) and the end ($\alpha = 1$) of the interpolation. This provides a weak lower bound on the number of polytope boundaries crossed during interpolation.

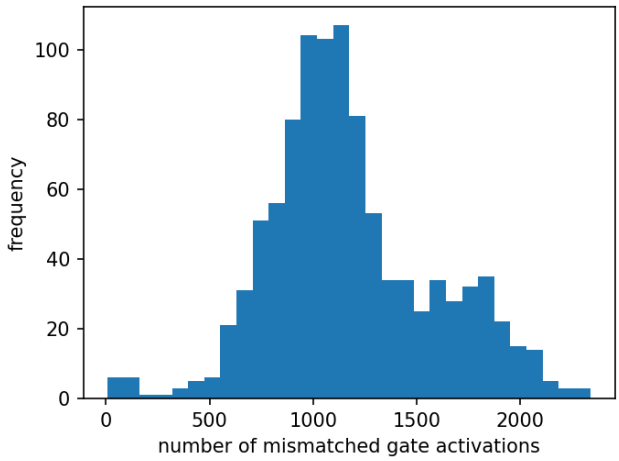

Figure 25: Histogram of the number of gate activations that switch signs during the interpolation in `Qwen2-0.5B`. Despite observing typically only one stable region boundary during interpolation, we see hundreds or thousands of gate activations changing signs, suggesting stable regions encompass many polytopes.

As shown in Figure 25, we typically observe hundreds or thousands of gates switching signs between two prompts, while our interpolation experiments suggest crossing only one stable region boundary. This provides direct evidence that stable regions are indeed much larger than individual polytopes.

G Relationship between output sensitivity and semantic similarity

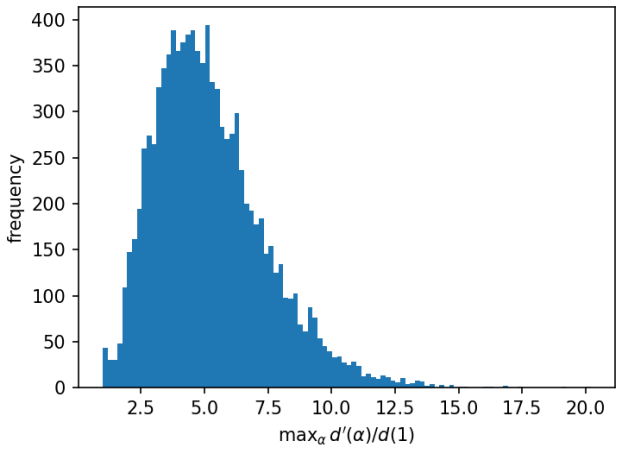

Figure 26: Distribution of maximum derivatives of the relative distance function in `Qwen2-0.5B`. The long right tail corresponds to prompt pairs that cross stable region boundaries.

To further investigate the relationship between semantic similarity and stable regions, we analyzed prompt pairs based on the maximum derivative of the relative distance ($\max_\alpha d'(\alpha)/d(1)$) in `Qwen2-0.5B`. We found that prompt pairs with low maximum derivative (< 1.2) consistently share the same last token, while pairs with high maximum derivative (> 10) have different last tokens. This analysis provides quantitative evidence for the relationship between semantic similarity and stable regions, supporting our earlier qualitative observations with manually selected prompts.

Here are representative examples:

G.1 Low sensitivity pairs (same last token)

p_A = “\r\n\r\nfor the little embellishments you speak of,”
p_B = “ckles’ arrangements with the\r\n\r\n Angel,”

p_A = “\r\n\r\n\r\nOak then struck up “Jockey to the”
p_B = “ nothing. The sale of big furniture barely paid the”

p_A = “ being out of the question, we did the\r\n\r\n”
p_B = “;\r\n\r\nAnd from about him fierce effusion rolled\r\n\r\n”

p_A = “\r\n\r\n it had scratched the varnish from the”
p_B = “ically; but shall be content\r\n\r\nto produce the”

p_A = “003:006 And when the”
p_B = “ a year since, how much I am in the”

p_A = “ilion\r\n\r\nand are beckoning to us.”\r\n\r\n\r\n”
p_B = “ answer; but the accent was decisive enough.\r\n\r\n\r\n”

p_A = “ I scattered them among the heathen, and”
p_B = “ show himself unto\r\n\r\nyou, my children, and”

p_A = “Israel, which he commanded our fathers, that they”
p_B = “ the LORD.\r\n33:31 And they”

p_A = “ quietly into the tree\r\n\r\nbeside him.\r\n\r\n\r\n”
p_B = “ choir, Thea saw that she must have\r\n\r\n”

p_A = “ be Jeddak of Helium. Say\r\n\r\n”
p_B = “ne to hit um wid a rock? How\r\n\r\n”

p_A = “ hide the city of New York under it, and”
p_B = “: and their glory, and their multitude, and”

p_A = “ subterranean\r\n\r\nchambers for ages,”
p_B = “, to Tarzan’s\r\n\r\nconsternation,”

p_A = “ to be\r\n\r\n entitled to his full name.”
p_B = “ He took up his candle \r\n\r\n\t\t\tto start.”

p_A = “ and there was the old man down the path a”
p_B = “as it used to be to keep from reading a”

p_A = “ more natural time, and I was once more in”
p_B = “.”\r\n\r\n\r\n“Well, there’s a good deal in”

p_A = “, as represented by Sir Percy Blakeney,”
p_B = “ a hundred paces,\r\n\r\nalmost without drawing breath,”

p_A = “ thee.\r\nPALAMON.\r\n\r\nNo more; the”
p_B = “ Neither boy spoke. If one moved, the”

p_A = “GONERIL.\r\n\r\nThis admiration, sir,”
p_B = “: Let us go forth to meet my people,”

p_A = “ have been here a number\r\n\r\n of times,”
p_B = “ had laboured to do: and, behold,”